The Impact of AI and ML on Cyber Security

Artificial intelligence and machine learning are changing the natural order of things—right from how we work and how the economy runs, to the nature of today's warfare, communications, privacy protection norms, etc. While their long-term impact remains uncertain, these technologies are a huge help to cyber security experts as they can be used to quickly identify and analyse possible attacks.

Artificial intelligence (AI) is the ability of a computer program or a machine to think, learn and act like a human being. It is also a field of study that tries to make computers *smart*. AI is a sub-field of computer science. The main goal of AI is to enable the development of computer systems that are able to do the things that humans do. AI involves the study of different methods for making computers behave as intelligently as people. It is the concept of making machines capable of performing tasks without human intervention, such as building smart machines.

Machine learning (ML) is a subset of AI and is based on the idea of writing computer algorithms that automatically upgrade themselves by discovering patterns in existing data, without being explicitly programmed. It is also used to automatically analyse the way interconnected systems work in order to detect cyber attacks and limit their damage. The entire processing of ML tools depends on data. The more data an algorithm obtains, the more accurate it becomes and thus, the more effective the results it delivers.

The role of machines in cyber security

Machine learning and artificial intelligence (AI) are being applied more broadly across industries and applications than ever before, as computing power, data collection and storage capabilities increase. From the cyber security perspective, this means new exploits and weaknesses can quickly be identified and analysed to help mitigate further attacks.

The perfect fit

Machines are much better and more cost-efficient than humans when it comes to handling huge amounts of data and performing routine tasks. This is exactly what the cyber security industry needs at the moment, especially with the large number of new threats appearing every day. Most of these new threats can easily be classified under existing families or familiar types of threats. In most cases, spending time looking at each new threat in detail would, in all probability, be a waste of time for a researcher or reverse engineer. Human classification, especially in bulk, will be